

DIGITAL PATHOLOGY AI SCREENING REPORT

Attention Augmented Residual Deep Learning Framework for Lymphoma Detection

Case Reference: CASE-4710E6BA	Date/Time: 2026-09-02 16:53:18.609485
Uploaded File: sample_histology.jpg	Confidence Threshold: 80%

Screening Status:	LYMPHOMA DETECTED
Classification Subtype:	Chronic Lymphocytic Leukemia / Small Lymphocytic Lymphoma
Model Confidence:	85.16%

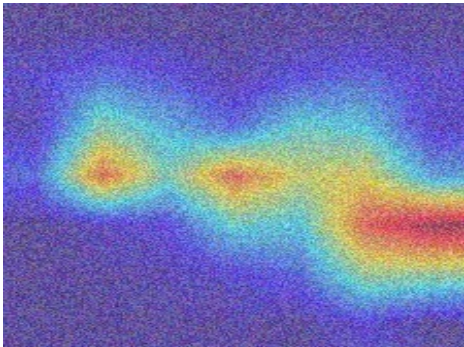
Three-Class Softmax Probability Distribution

Class / Subtype	Full Medical Designation	Calculated Probability
CLL	Chronic Lymphocytic Leukemia / Small Lymphocytic Lymphoma	85.16%
FL	Follicular Lymphoma	3.65%
MCL	Mantle Cell Lymphoma	11.19%

Visual Explainability: Attention Heatmap (Grad-CAM++)



Original Histological Slide



Grad-CAM++ Cellular Attention

Histopathological Morphology Summary

Chronic Lymphocytic Leukemia / Small Lymphocytic Lymphoma (CLL/SLL): Characterized by a diffuse proliferation of small, mature-appearing, monotonous round lymphocytes with dense clumped chromatin, indistinct nucleoli, and scant cytoplasm. Proliferation centers (pseudofollicles) containing prolymphocytes and para-immunoblasts may be present.

ACADEMIC RESEARCH & CLINICAL DECISION SUPPORT NOTICE:

This automated report was produced by the 'Attention Augmented Residual Deep Learning Framework for Lymphoma Detection' as a research-grade computer-aided screening tool. This system does not constitute a definitive medical or pathological diagnosis. Rejection or 'No Lymphoma Detected' indicates that the image features fell below the model's high-confidence threshold and must not be interpreted as

definitive proof of cancer-free tissue. All digital pathology findings must be correlated with clinical context, immunohistochemistry (CD5, CD10, CD20, Cyclin D1), and board-certified pathological review.